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	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC	BD	BE	BF	BG	BH	BI	BJ	BK	BL	BM	BN	BO	BP	BQ	BR	BS	BT	BU	BV	BW	BX	BY	BZ	CA	CB	CC	CD	CE	CF	CG	CH	CI	CJ	CK	CL	CM	CN	CO	CP	CQ	CR	CS	CT	CU	CV	CW	CX	CY	CZ	DA	DB	DC	DD	DE	DF	DG	DH	DI	DJ	DK	DL	DM	DN	DO	DP	DQ	DR	DS	DT	DU	DV	DW	DX	DY	DZ	EA	EB	EC	ED	EE	EF	EG	EH	EI	EJ	EK	EL	EM	EN	EO	EP	EQ	ER	ES	ET	EU	EV	EW	EX	EY	EZ	FA	FB	FC	FD	FE	FF	FG	FH	FI	FJ	FK	FL	FM	FN	FO	FP	FQ	FR	FS	FT	FU	FV	FW	FX	FY	FZ	GA	GB	GC	GD	GE	GF	GG	GH	GI	GJ	GK	GL	GM	GN	GO	GP	GQ	GR	GS	GT	GU	GV	GW	GX	GY	GZ	HA	HB	HC	HD	HE	HF	HG	HH	HI	HJ	HK	HL	HM	HN	HO	HP	HQ	HR	HS	HT	HU	HV	HW	HX	HY	HZ	IA	IB	IC	ID	IE	IF	IG	IH	II	IJ	IK	IL	IM	IN	IO	IP	IQ	IR	IS	IT	IU	IV	IW	IX	IY	IZ	JA	JB	JC	JD	JE	JF	JG	JH	JI	IJ	JK	JL	JM	JN	JO	JP	JQ	JR	JS	JT	JU	JV	JW	JX	JY	JZ	KA	KB	KC	KD	KE	KF	KG	KH	KI	KJ	KK	KL	KM	KN	KO	KP	KQ	KR	KS	KT	KU	KV	KW	KX	KY	KZ	LA	LB	LC	LD	LE	LF	LG	LH	LI	LJ	LK	LL	LM	LN	LO	LP	LQ	LR	LS	LT	LU	LV	LW	LX	LY	LZ	MA	MB	MC	MD	ME	MF	MG	MH	MI	MJ	MK	ML	MM	MN	MO	MP	MQ	MR	MS	MT	MU	MV	MW	MX	MY	MZ	NA	NB	NC	ND	NE	NF	NG	NH	NI	NJ	NK	NL	NM	NN	NO	NP	NQ	NR	NS	NT	NU	NV	NW	NX	NY	NZ	OA	OB	OC	OD	OE	OF	OG	OH	OI	OJ	OK	OL	OM	ON	OO	OP	OQ	OR	OS	OT	OU	OV	OW	OX	OY	OZ	PA	PB	PC	PD	PE	PF	PG	PH	PI	PJ	PK	PL	PM	PN	PO	PP	PQ	PR	PS	PT	PU	PV	PW	PX	PY	PZ	QA	QB	QC	QD	QE	QF	QG	QH	QI	QJ	QK	QL	QM	QN	QO	QP	QQ	QR	QS	QT	QU	QV	QW	QX	QY	QZ	RA	RB	RC	RD	RE	RF	RG	RH	RI	RJ	RK	RL	RM	RN	RO	RP	RQ	RR	RS	RT	RU	RV	RW	RX	RY	RZ	SA	SB	SC	SD	SE	SF	SG	SH	SI	SJ	SK	SL	SM	SN	SO	SP	SQ	SR	SS	ST	SU	SV	SW	SX	SY	SZ	TA	TB	TC	TD	TE	TF	TG	TH	TI	TJ	TK	TL	TM	TN	TO	TP	TQ	TR	TS	TT	TU	TV	TW	TX	TY	TZ	UA	UB	UC	UD	UE	UF	UG	UH	UI	UJ	UK	UL	UM	UN	UO	UP	UQ	UR	US	UT	UU	UV	UW	UX	UY	UZ	VA	VB	VC	VD	VE	VF	VG	VH	VI	VJ	VK	VL	VM	VN	VO	VP	VQ	VR	VS	VT	VU	VV	VW	VX	VY	VZ	WA	WB	WC	WD	WE	WF	WG	WH	WI	WJ	WK	WL	WM	WN	WO	WP	WQ	WR	WS	WT	WU	WV	WW	WX	WY	WZ	XA	XB	XC	XD	XE	XF	XG	XH	XI	XJ	XK	XL	XM	XN	XO	XP	XQ	XR	XS	XT	XU	XV	XW	XX	XY	XZ	YA	YB	YC	YD	YE	YF	YG	YH	YI	YJ	YK	YL	YM	YN	YO	YP	YQ	YR	YS	YT	YU	YV	YW	YX	YZ	ZA	ZB	ZC	ZD	ZE	ZF	ZG	ZH	ZI	ZJ	ZK	ZL	ZM	ZN	ZO	ZP	ZQ	ZR	ZS	ZT	ZU	ZV	ZW	ZX	ZY	ZZ
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Fig.1. Datatable of the users' log

```
{idrecord:2984,
encode:length,
dimension:low,
task_id:0,
case_id:1,
correct_value:1,
user_id:F000,
user_input:1,
timestamp:1651371445820,
mousetrack:[{'x':643,'y':367,'time':1651371445180,'baseLength':323.4},{'x':627,'y':353,'time':1651371
445213,'baseLength':323.4}],
comments:click axis,
vertices:0,
innerline:0,
first_write:null,
is_correct:True}
```

2. Interface of the workshop system

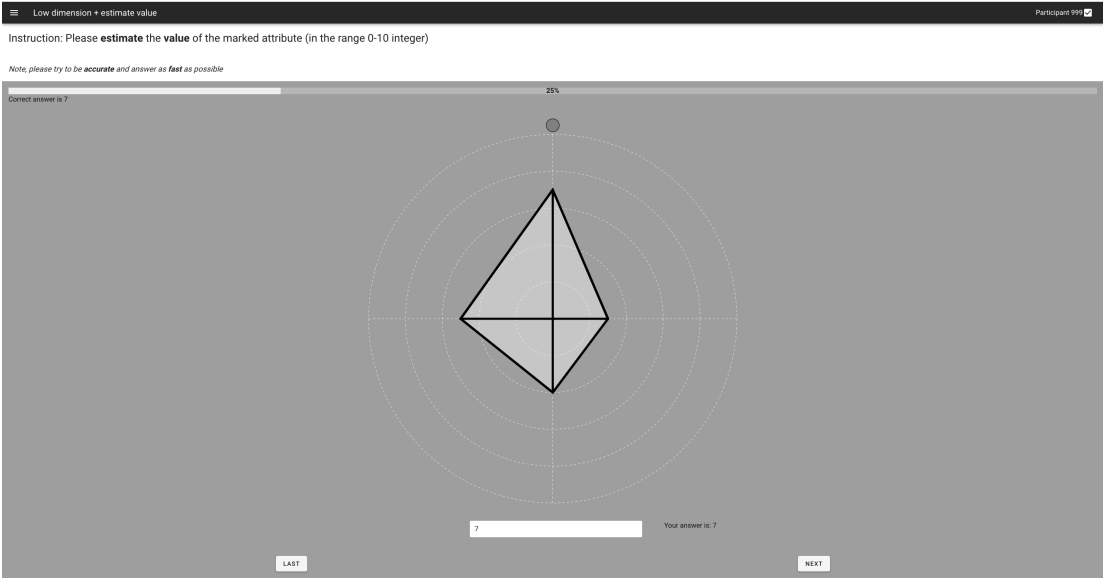


Fig.2. Interface of the workshop system: Estimate the value (Tutorial)

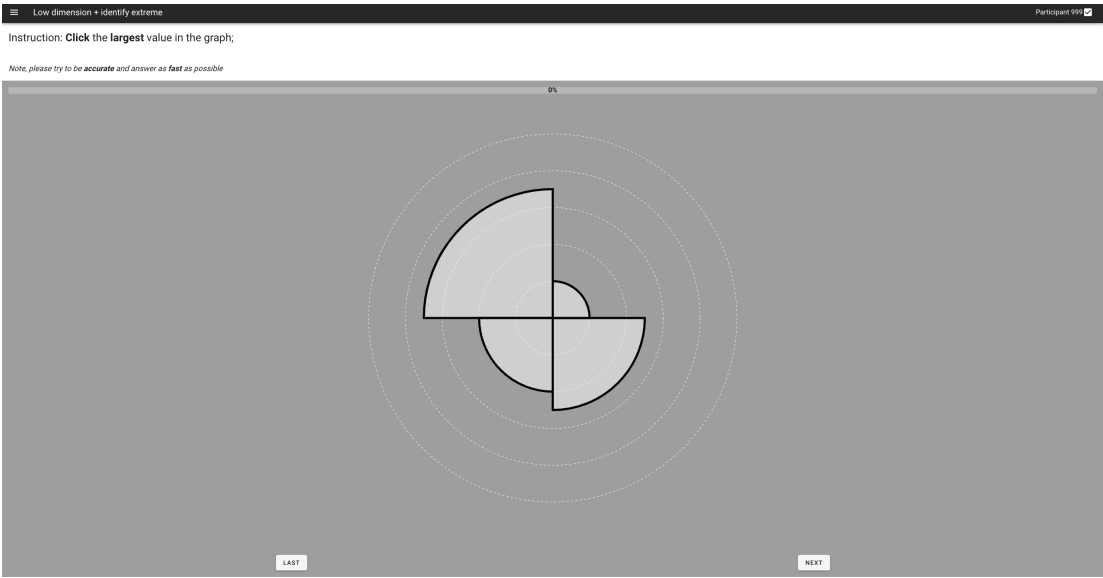


Fig.3. Interface of the workshop system: Finding extremes

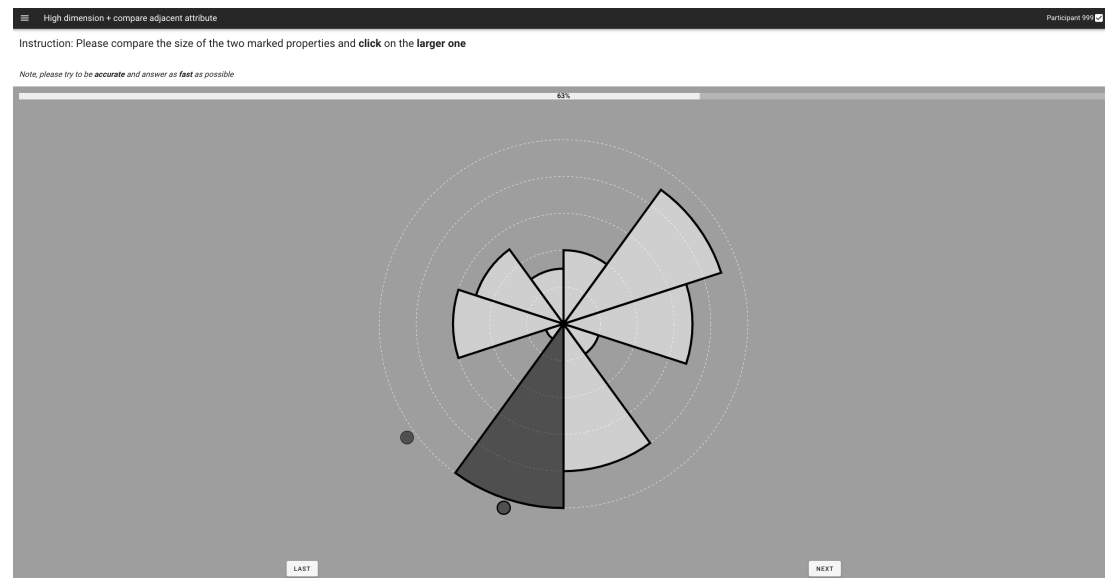


Fig.4. Interface of the workshop system: Comparing values of adjacent attributes (Answered)

3. The questionnaire example for user study

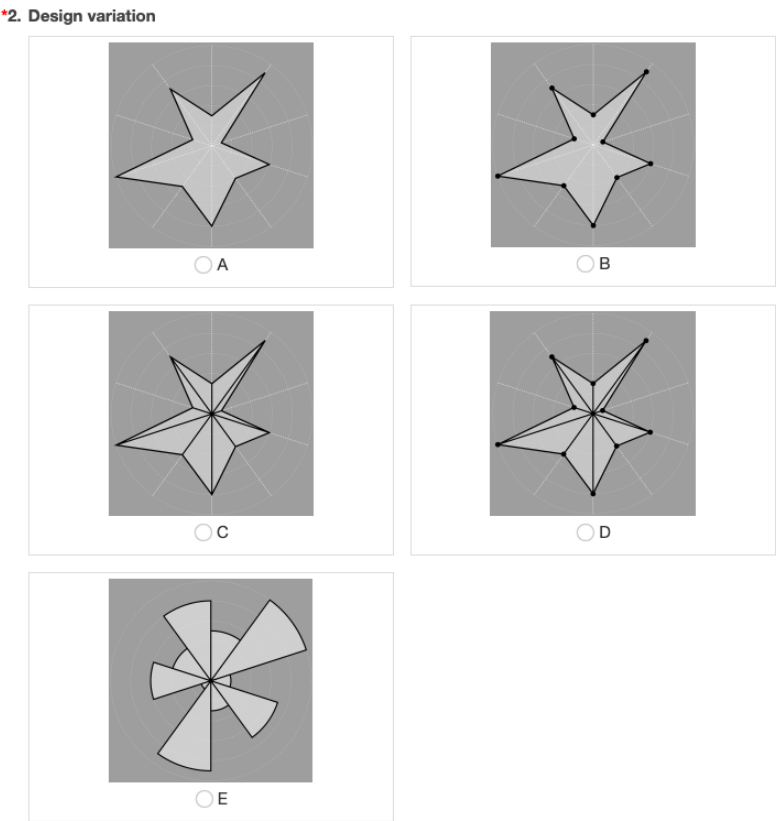


Fig.5. Questionnaire - 1

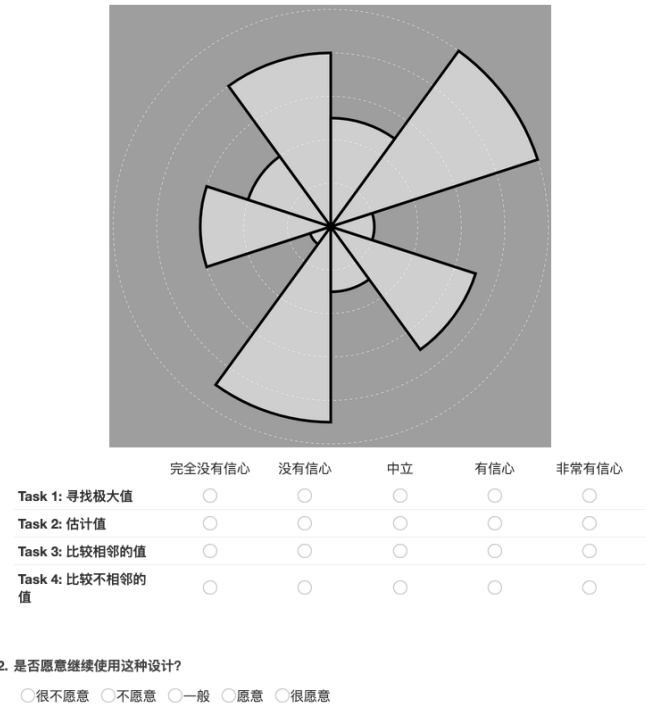


Fig.6. Questionnaire - 2

4. Original results of the preference

Table.1. The original results of the user's preference

id	1(Most like)	2	3	4	5(Most dislike)
1	A	E	B	C	D
2	B	E	A	D	C
3	C	E	A	D	B
4	D	E	C	B	A
5	D	E	B	C	A
6	E	A	B	C	D
7	E	A	C	B	D
8	E	A	B	C	D
9	E	B	A	C	D
10	E	B	A	C	D
11	E	B	A	D	C
12	E	B	D	A	C
13	E	B	D	C	A

14	E	B	D	C	A
15	E	C	B	A	D
16	E	C	D	A	B
17	E	C	D	B	A
18	E	C	D	A	B
19	E	D	B	C	A
20	E	D	C	B	A
21	E	D	C	B	A
22	E	D	C	B	A
23	E	D	C	B	A
24	E	ABCD			
25	E	BD	AC		

5. Samples of the Comments of the designs

The comments selected from no.1 - no.5 as followed:

Participant 1:

AB is more messy, feel no focus. A compares the two prominent sharp points, while B is relatively better. The lines of C and D are marked, and they are more clearly compared, especially if the final value of the figure is indicated at the point. Because the four tasks are the location of interest, CD is easier to see than other ones, because the area is more obvious. Valuation will be stunned, no point to indicate clear.

Participant 2:

Because when I saw the B, I was directly looking for that point. It's just the more striking. Then I can re-point the corresponding position and then correspond to its value.

E is not very good when it is not adjacent, when the difference is 1, it is slow to see the difference; no B C will attract attention A to compare the adjacent values, judging the slope of the line, and judging the position of the nonadjacent values, take more time.

Participant 3:

E: When looking for the maximum value, the more the outer circle arc, the longer, the larger the area, the faster the choice. Area, arc is all considered. With the length diagram, the more out it may be, the arc may be or not pointed. It needs to consider that the two adjacent values of D are too messy, and the line of line C may be useful (although not felt), more obvious than the point. But if it is not adjacent, the reference value of the line is not high if the auxiliary line is removed, E can also choose the right, C is very difficult.

Participant 4:

AB can not find the center of the circle, and the center may be biased to the geometric center, resulting in a deviation. The tip of C does not determine the position (vertex helps, intuitive) cable to see the length of the line, no line to see the scale; the valuation scale, compare the size to see the

length.

Participant 5:

E find the fastest, the data size through the area, arc length, radius, more intuitive CD, points plus no feeling difference

C has the inline line to see more clearly, with blocks

AB is similar, if the difference is more obvious, look at the line length; if the difference is very small, look at the vertex position to estimate the specific value